



OpenSCAD Home Models

OpenSCAD Home Models

May 2020

A quick project where I modeled several pieces of furniture from my parents' house in OpenSCAD — around the time I got my first 3D printer (and had plenty of time thanks to COVID).

I wanted to try out parametric modeling and thought our house's furniture would be a good subject. My mom also got into it, painting the real pieces while I modeled them, which made it a fun shared project.

Each model was built from scratch using real measurements and a bit of creative freedom, especially for the sofas. The collection includes desks, shelves, cabinets, and other everyday furniture.

All models are available on [GitHub](#).

Notes:

- Modeled entirely in OpenSCAD
- Parametric dimensions for easy scaling and modification
- Based on real furniture measurements and photos
- Some artistic license for softer shapes like sofas

OpenSCAD Home Models

May 2020 / 3D Modeling, Parametric Design

A collection of real-world furniture pieces from my parents' house, modeled from scratch with parametric dimensions using OpenSCAD.

(This is the perfect spot for a gallery or a collage of rendered images of your 3D models.)

The Story

This was a quick project I took on around the time I got my first 3D printer (and

had plenty of time thanks to the COVID lockdown). I wanted to learn parametric modeling, and I thought the furniture in our house would be a great subject. It turned into a fun shared project with my mom, who started painting the real pieces while I was modeling their digital counterparts. Each model was built from scratch using real measurements, though I took some creative freedom for softer shapes like the sofas. The collection includes desks, shelves, cabinets, and other everyday furniture.

Key Features

- **Fully Parametric:** All models are built with parametric dimensions, allowing for easy scaling and modification.
- **Code-Based Modeling:** Modeled entirely in OpenSCAD, representing physical objects through code.
- **Based on Real Objects:** Dimensions and designs are based on real-world furniture measurements and photos.
- **Diverse Collection:** Includes a variety of common household furniture like desks, shelves, and cabinets.

Tech Stack

- **Primary Software:** OpenSCAD
- **Concepts:** Parametric Modeling, 3D Design

Challenges & Lessons Learned

The main challenge was accurately representing complex and soft shapes, like sofas, using a purely programmatic CAD tool like OpenSCAD. This required creative problem-solving to approximate organic forms with geometric primitives. This project was an excellent introduction to the power and precision of parametric design and the unique workflow of "programming" physical objects.

Links

- [GitHub Repository](#)